A SOCIOECONOMIC ANALYSIS OF U.S. STATES

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Data Acquisition and Loading

Tool Used: Microsoft Excel

Dataset Source and Description

This project utilizes a publicly available dataset from Kaggle that explores the relationship between education levels, household income, and industry employment distribution across U.S. Congressional Districts during 2020 and 2021. The dataset comprises three structured CSV files:

- Education.csv – Educational attainment rates by district and year
- Finance.csv – Household income distribution across income brackets
- Industry.csv – Employment across industries, disaggregated by gender

This dataset enables a rich, multidimensional analysis of socioeconomic trends across regions.

Source:

Kaggle Dataset: https://www.kaggle.com/datasets/mittvin/u-s-census-dataset-educationfinance-industry/data

Reasons for Dataset Selection:

We selected this dataset for the following key reasons:

- **Relevant and timely**, covering key issues like income inequality, education access, and workforce trends.
- **Multi-domain coverage** (education, income, industry) supports deeper cross-sectional analysis.
- **Rich granularity**, with data from 437 districts over two years and gender-wise breakdowns
- **Easy to integrate**, with consistent keys (cd and Year) across all files.
- **Well-suited for visualisation**, enabling bar charts, geo maps, and trend comparisons for impactful BI storytelling.

Problem-Solving Potential:

This project reveals important socioeconomic patterns by linking education, income, and employment data. Key insights include:

- **Regional inequality**: District-level data helps identify areas needing policy support and resource allocation.
- **Education-income correlation**: Supports strategic decisions for funding and workforce planning.
- **Gender disparities**: Highlights gaps in industry employment, aiding gender equity initiatives.
- **State benchmarking**: Enables cross-state comparisons to guide best practices and track progress.

Integration Approach using Excel:

To build a unified dataset:

1. Filtered all files to retain only the years 2020 and 2021 to ensure consistency, as the Education file lacked 2019 data.
2. A common join key 'Concat' was created by concatenating Year and cd in excel.
3. Match accuracy was verified using Excel's 'MATCH' function.
4. 'XLOOKUP' was used to merge all columns from Finance and Industry into the Education sheet.
5. Formatting inconsistencies were resolved using 'TRIM, CLEAN, and LOWER.'
6. Additional columns were appended, by mapping full state names from the cd values.
7. The final integrated dataset serves as the analytical foundation for subsequent BI tasks.

Data Analysis

Tool Used: Python (Google Collab)

A. Data Exploration, Profiling, and Cleaning

Establishing the dataset's relevance and structure was necessary to ensure it could effectively support the project's analysis and reporting goals. Profiling began with identifying key attributes that impact income distribution, education levels, and sector-wise employment. These attributes were then systematically described to uncover data structure, formatting issues, and reliability.

1. A comprehensive **data dictionary** was created for over 15 key columns, including time (Year), geography (State Name), education levels, income brackets, and employment sectors. The dictionary captured:
   - Column names (Year, State Code, State Name) • Data types (int64, object).
   - Format or units (e.g., percentages, currency).
   - Field sizes and sample values.
   - Keyword tags such as "Time", "Location", or "Industry".

	Column Name	Data Type	Sample Value	Format	Field Size	Keywords	Description
0	Year	int64	2020	Integer	4	Time	The year the data was recorded (typically Cens...
1	cd	object	0_AK	Text	4	Location	Census code representing the state and year co...
2	State Code	object	AK	Text	2	Location	Abbreviated state code (e.g., CA for California).
3	State Name	object	Alaska	Text	6	Location	Full name of the U.S. state.
4	Conccat	object	20200_AK	Text	8	ID	Concatenated key field (State + Year).
5	Bachelors_degree_or_higher	float64	121098.0	Decimal	8	Education	Population with a bachelor's degree or higher.
6	high_school_or_some_degree	int64	309698	Integer	6	Education	Population with at least high school or some c...
7	Less_than_$5000	int64	5949	Integer	4	Income	Individuals with income less than $5,000.
8	$150000_or_more	int64	47139	Integer	5	Income	Individuals with income greater than or equal ...
9	Total_Agriculture_forestry_fishing_hunting_mining	int64	9356	Integer	4	Industry	Total employment in agriculture, forestry, fis...
10	Total_Transportation_warehousing_utilities	int64	21114	Integer	5	Industry	Total employment in transportation, warehousin...
11	Total_Finance_insurance_realestate_rental_leasing	float64	10075.0	Decimal	7	Industry	Total employment in finance, insurance, real e...
12	Total_Professional_scientific_management_admin...	int64	18855	Integer	5	Industry	Total employment in professional, scientific, ...
13	Total_Educationalservices_healthcare_socialass...	int64	52808	Integer	5	Industry	Total employment in education, healthcare, and...
14	Total_Public_administration	int64	32793	Integer	5	Industry	Total employment in public administration.

Fig 2.1: Structured Data Dictionary
Source: Google Collab File

2. To further understand the summary of the central tendency, dispersion, and shape of a dataset's distribution, **Descriptive statistics** were computed using ".describe()". This revealed the following patterns and anomalies across attributes:
   - For Column "Less_than_$5000" income, the mean is ~9105 while the median is ~8136, suggesting a right-skewed distribution.
   - The mean value for "Bachelors_degree_or_higher" is around 145,152 with a high standard deviation (~80,654), indicating significant variation in educational attainment across regions.
   - A frequency plot of State Name revealed an imbalance in state representation.

	Year	Bachelors_degree_or_higher	high_school_or_some_degree	Less_than_high_school_graduate	Less_than_$5000
count	887.0	840.0	887.0	887.0	887.0
mean	2020.499436	145151.992857	269343.59301	49396.314543	9105.121759
std	0.500282	80653.993091	108925.100091	30542.117827	9061.474225
min	2020.0	2872.0	12603.0	2072.0	64.0
25%	2020.0	91828.75	198717.0	31724.5	5611.5
50%	2020.0	125764.0	266605.0	43688.0	8136.0
75%	2021.0	180791.75	340419.5	58328.0	11039.0
max	2021.0	561182.0	1191468.0	257422.0	178865.0

8 rows × 56 columns

Fig 2.2: Descriptive Statistics
Source: Google Collab File

These trends are especially important to **policy makers** and **economists** for monitoring disparities in education and income across regions.

3. **Missing value analysis** used the command
df.isnull().sum().plot(kind='bar'), confirming the existence of few null values. Fields such as "Total_Construction", etc. had missing entries. The values were imputation using respective column's mean to preserve distribution and consistency.

	Missing Values	Percentage (%)
Bachelors_degree_or_higher	47	5.298760
$10000_to_$14999	47	5.298760
$50000_to_$74999	47	5.298760
Total_Construction	46	5.186020
Total_Finance_insurance_realestate_rental_leasing	30	3.382187
Male_Construction	30	3.382187
Male_Finance_insurance_realestate_rental_leasing	30	3.382187
Female_Educationalservices_healthcare_socialassistance	30	3.382187

Fig 2.3: Numerical Summary for Missing Value Source:
Google Collab File

Fig 2.4: Extract of the Bar plot For missing values
Source: Google Collab File

4. **Outlier detection** involved both visual inspection using Boxplots and statistical method of Inter-Quartile Range:
   - The boxplots mostly exposed outliers in income categories like “Less_than_$5000”, etc.
   - The **IQR (Interquartile Range)** method was applied to cap outliers without removal.
   - Visual comparisons before and after capping showed smoother, more interpretable distributions.

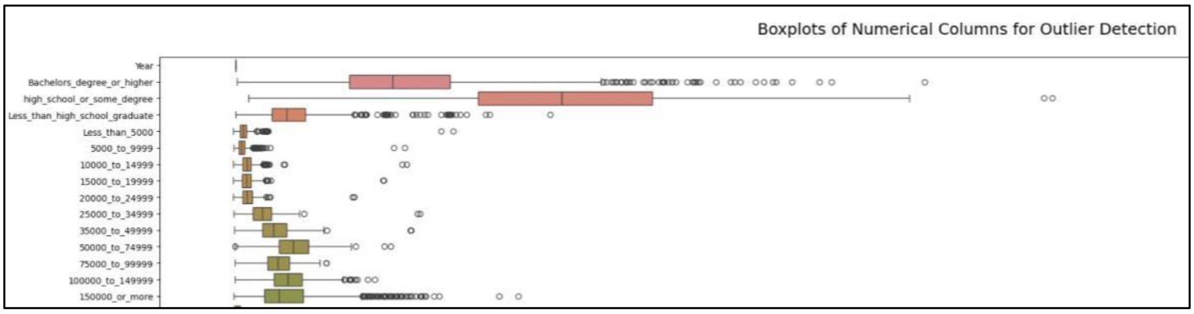


Fig 2.5: Extract of Boxplot outliers before applying IQR capping
Source: Google Collab File

Fig 2.6: Boxplot after applying IQR capping
Source: Google Collab File

5. **Data quality and formatting** issues were addressed in the following manner:
   - Removed 13 duplicate rows using “df.drop_duplicates()”.
   - Standardized categorical values (e.g., State Name) using “.str.strip()” and “.str.title()” to eliminate spacing and casing inconsistencies.
   - Corrected column data types with “.astype()” to ensure compatibility with BI tools.

B. Data Transformation and Wrangling

To enhance data quality, efficiency, and analytical relevance, the dataset underwent several transformation steps:

- **Dropped 16 irrelevant columns** using "df.drop()" due to redundancy, low variance, or lack of alignment with stakeholder goals.
- **Renamed lengthy columns** such as "Female_Arts_entertainment_recreation_accommodation_foods ervices" to "Female_Arts" for clarity in visuals and reports.
- **Grouped income brackets** into broader ranges aligned with formats used in policy and economic reporting.
- **Verified and corrected data types** to match requirements of tools like Power BI and Excel.

All steps were executed in **Python** using **Pandas** and **Seaborn**. The cleaned dataset was exported via df.to_csv('cleaned_dataset.csv', index=False) and was now structured for BI visualization and reporting.

This structured approach ensured that the final dataset is accurate, stakeholder-friendly, and ready to support **evidence-based decision-making** by **educational leaders**, **economists**, and **government analysts**.

Data Presentation and Interpretation

Tool Used: Tableau

Visualisation 1: Gender-wise Employment by Industry

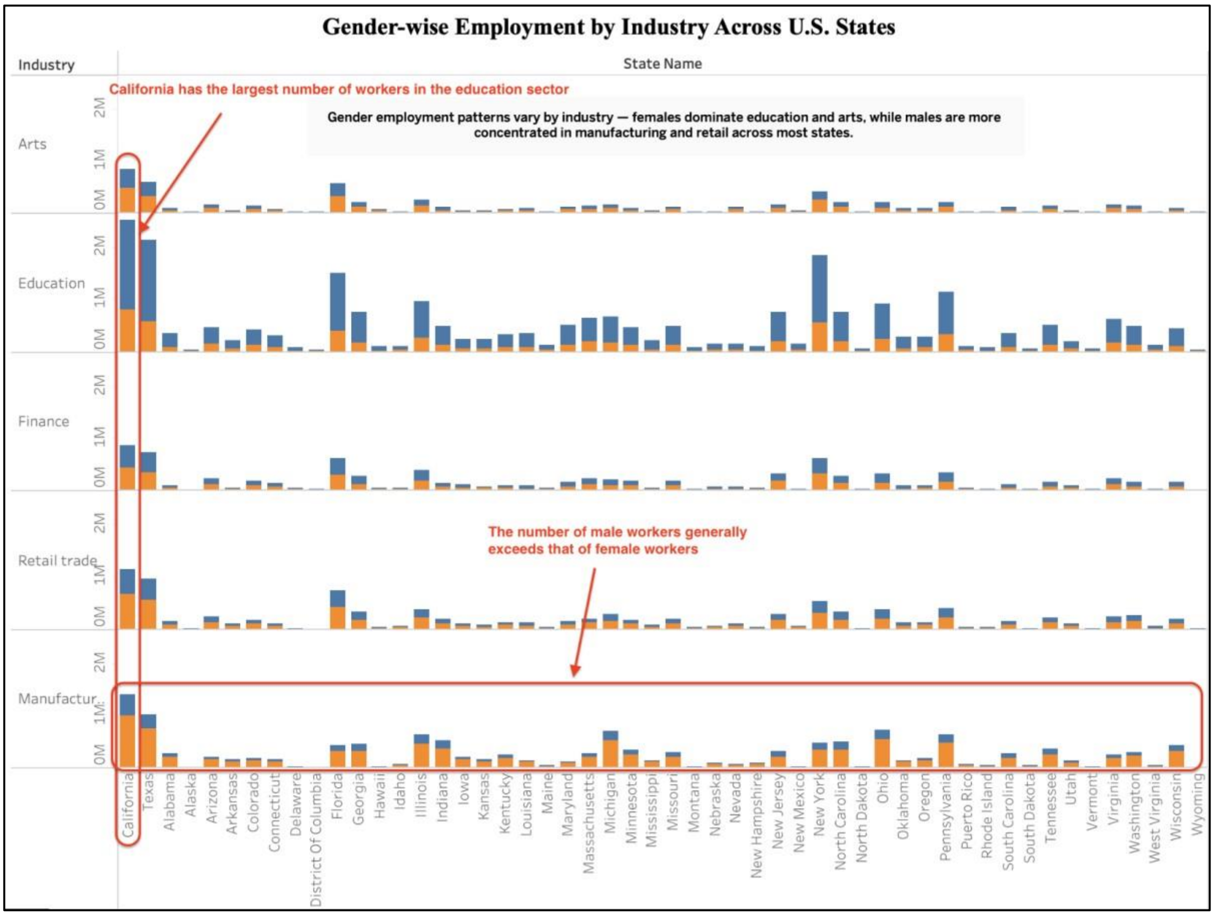


Fig 3.1: Gender Wise- Bar Chart
Source: Tableau File

Why This Chart Was Chosen:

A horizontal stacked bar chart is ideal for showing categorical comparisons across two dimensions—industry and gender—while maintaining clarity over multiple states. This format allows us to:

- Highlight distinct gender-based trends in employment across industries
- Identify industries with noticeable gender imbalance (e.g., male dominance in Manufacturing, female dominance in Education)
- Provide policymakers with evidence to design gender equity programs or targeted workforce development strategies

The chart uses a clean layout, consistent color scheme, and annotations to ensure insights are accessible even to non-technical audiences.

Interpretation:

- **Target Audience:** Labor economists, state workforce boards, and gender equity policymakers tracking industry-based gender trends.
- It shows female dominance in education and arts sectors**,** while males are highly concentrated in manufacturing and retail trade—particularly in states like California, Texas, and New York**.**
- In most industries, the **number of male workers exceeds female workers**, especially in manufacturing, indicating ongoing gender imbalances.
- This highlights sector-specific gender gaps, enabling analysis of occupational segregation and potential workforce inequality.
- **Actionable Insight**: Informs gender-targeted employment programs, diversity hiring policies, and industry-specific training initiatives to promote balance.
- **Real-world Impact**: Supports strategic planning for inclusive economic growth, ensuring both genders have equal access to high-employment industries across states.

Visualisation 2: Scatter Plot – Education Level vs High-Income Households

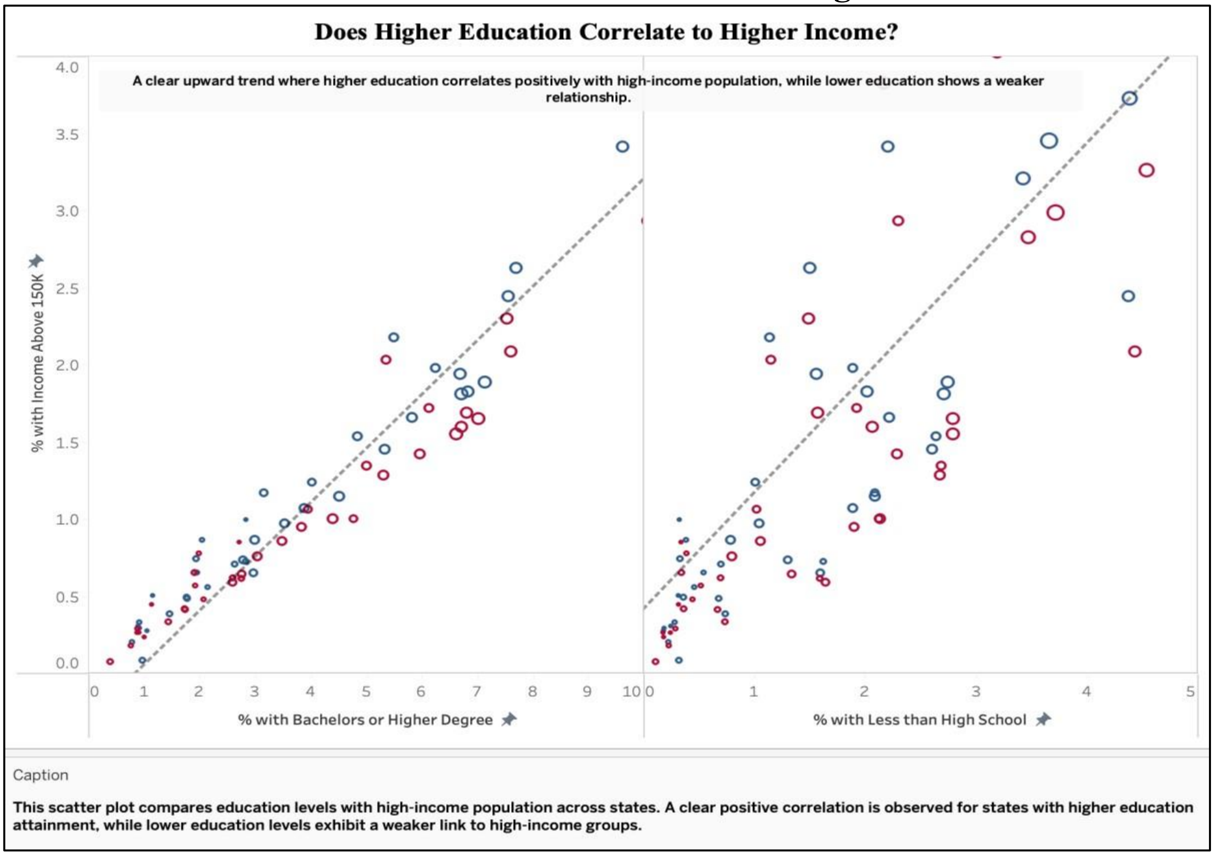


Fig 3.2: Education Vs Income Scatter Plot
Source: Tableau File

Why This Chart Was Chosen:

A scatter plot is effective for showing the relationship between two continuous variables. This dual format allows us to:

- Clearly demonstrate the positive correlation between higher education and high income

- Highlight that some states still achieve high income levels without high education attainment, revealing nonlinear socioeconomic factors

This kind of visual insight is essential for your **audience** (policy makers and educators), as it allows them to:

- Justify funding toward educational initiatives
- Identify states where **alternative economic drivers** may influence income (e.g., tech, oil, or manufacturing.

The visual uses dual panels, clear axis labels, and consistent colors to support interpretability without requiring technical knowledge.

Interpretation:

- Tailored for **education policymakers and economic strategists** assessing the impact of education on income distribution.
- The left panel shows a **strong positive correlation** between higher education (% with bachelor's or higher) and high-income population—states with more educated populations tend to have more residents earning above $150K.
- The right panel reveals **a** weaker or scattered trend between low education levels and high income, suggesting limited upward mobility for less-educated populations.
- This highlights **clear disparities**, higher education correlates with economic advantage, reinforcing the role of education in income mobility.
- **Actionable Insight:** Justifies increased investment in higher education**,** scholarship programs**,** and degree accessibility initiatives, especially in underperforming states.
- **Real-world Impact:** Enables decision-makers to align economic development with education reform, ensuring policies support income growth through academic advancement.

Visualisation 3: Geographic Heatmap – State-wise Income Distribution

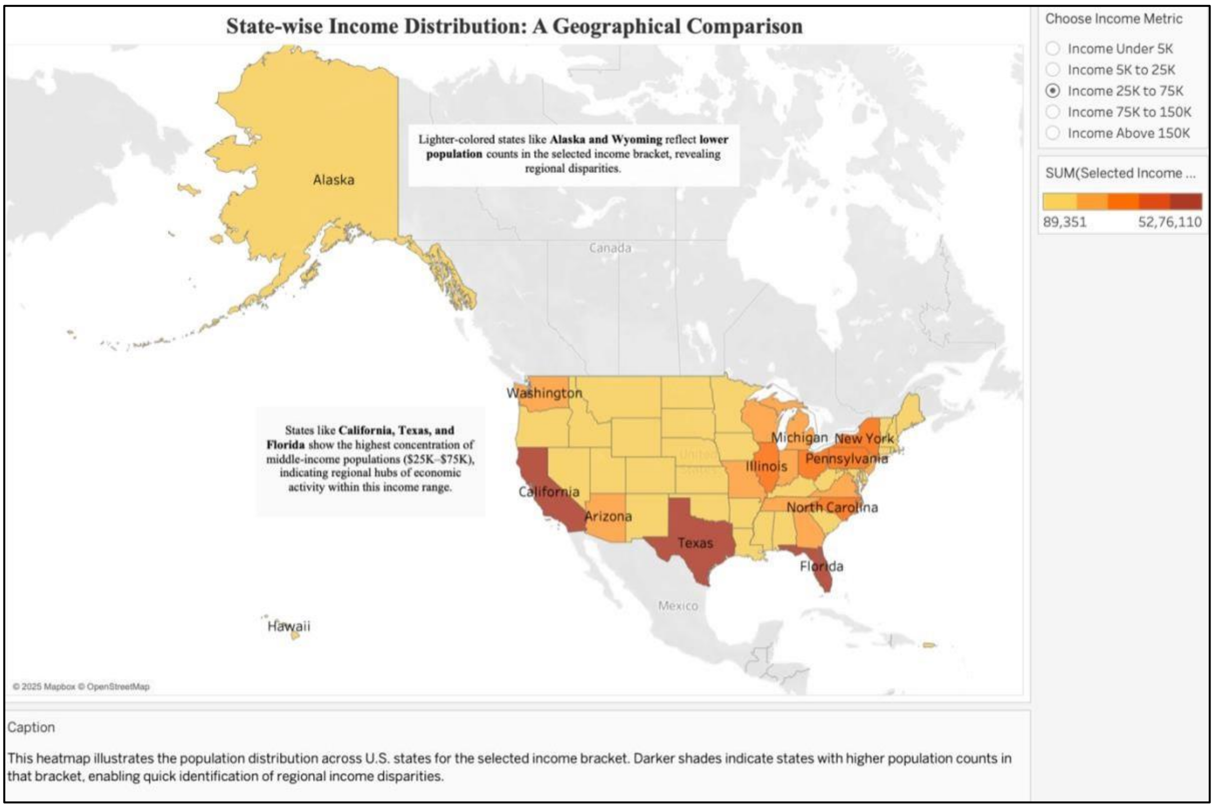


Fig 3.3: Geographic Heat Map
Source: Tableau File

Why This Chart Was Chosen:

A geographic heatmap was selected for its ability to communicate spatial data patterns intuitively. It allows stakeholders to:

- Visually compare state-wise income distributions at a glance
- Pinpoint regional hubs of middle-income populations, useful for federal aid, education planning, or workforce programs
- Detect states with lower economic density in this range, enabling focused policy responses for income support

The visual includes a clear title, interactive income bracket filters, and contextual annotations to enhance interpretability for decision-makers. This map is especially relevant for economic planners and regional development authorities.

Interpretation:

- This heatmap is designed for **government agencies and policymakers** to assess the distribution of middle-income households ($25K–$75K) across U.S. states.
- The map reveals concentration of middle-income populations in states like Texas, California, and Florida, suggesting these as key economic hubs for this income group.

- **Regional Disparities:** Lighter-shaded states such as Alaska and Wyoming reflect lower population counts in this bracket, highlighting income distribution gaps that may require targeted interventions.
- **Actionable Insight:** Enables strategic decisions such as adjusting federal funding models, allocating housing/education grants, or launching state-specific workforce programs.
- **Real-world Impact:** Helps government bodies prioritise economic aid, monitor progress on income equality goals, and design location-sensitive policies.

Visualisation 4: Treemap – Population Distribution by Education Level Across States

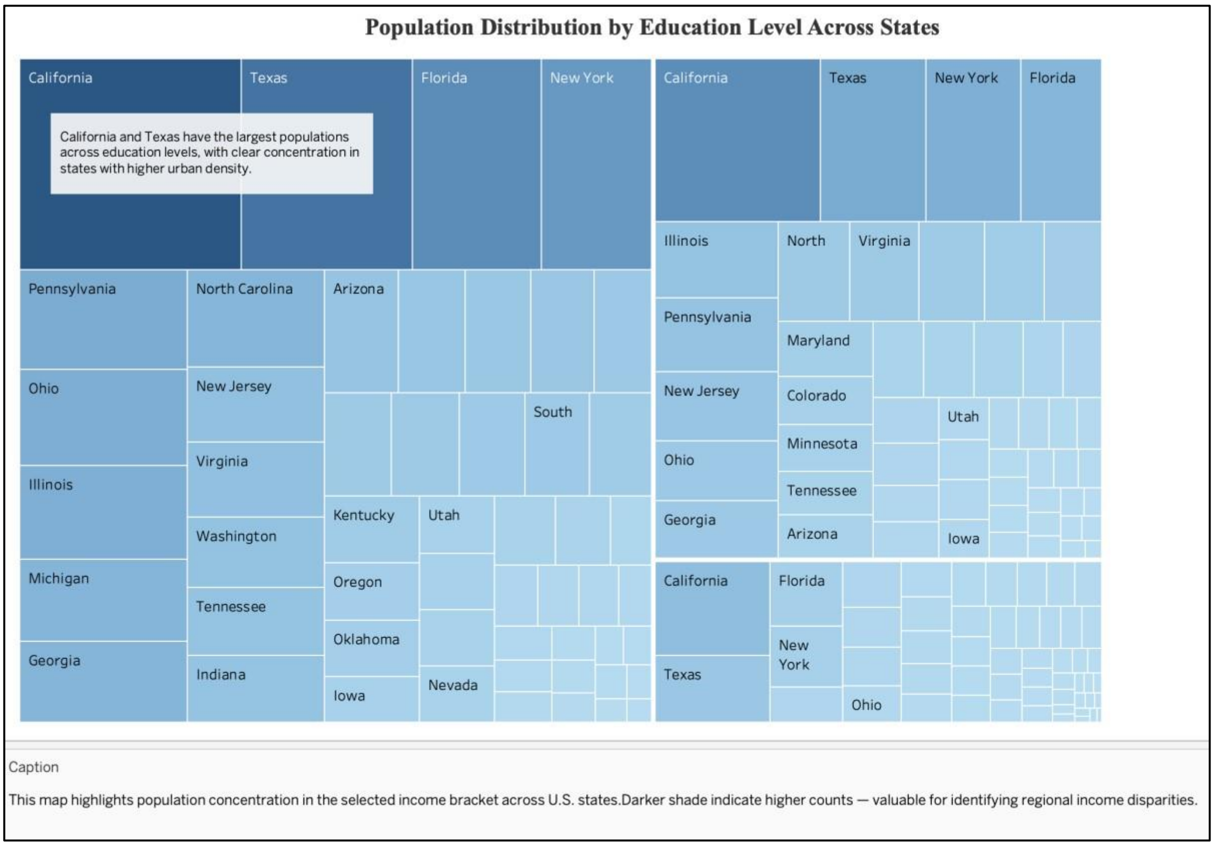


Fig 3.4: Education Treemap
Source: Tableau File

Why This Chart Was Chosen:

A treemap is effective for displaying hierarchical and proportional data, making it easier to compare states based on population scale. It was selected to:

- Highlight how education-related population sizes vary by geography
- Emphasize regional density, aiding policymakers in identifying areas that may require more education infrastructure or targeted investment
- Offer a visually compact and intuitive way to interpret population concentration across all 50 states

This visual insight is particularly relevant to **urban planning, resource allocation, and educational policy**. By clearly showing where the largest education populations reside, planners and educators can focus their initiatives where they'll have the most impact.

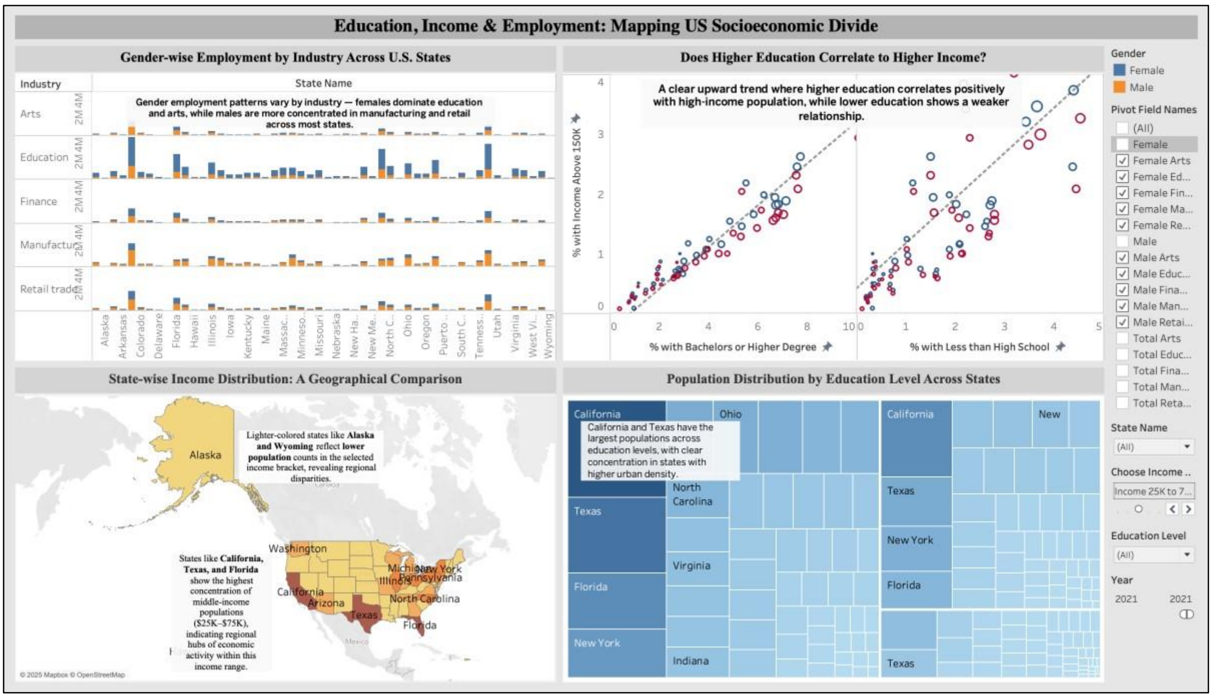
The use of consistent colour gradients and readable labels makes this chart easy to interpret even for non-technical stakeholders.

Interpretation:

- Target Audience: **State Education policymakers**
- The treemap reveals high education-level populations in states like California, Texas, and Florida, aligning with urban density and greater institutional access.
- **Regional Disparities:** Smaller, lighter-shaded states such as Iowa and Nevada reflect lower education-level population counts, indicating potential gaps in education reach or population size.

  This enables quick state-level comparisons, helping identify where education demand is concentrated.
- **Actionable Insight:** Supports decisions like expanding school funding, opening new institutions, or launching state-specific education support programs.
- **Real-world Impact:** Equips policymakers to prioritise underserved states, improve education accessibility, and track progress on national equity goals.

Static Dashboard



The dashboard is designed to offer a cohesive and visually intuitive overview of patterns across education, income, and employment.

- Positioned top-left, the stacked bar chart highlights gender disparities in industry employment, immediately drawing attention to sectors like **education and arts** where **female** representation dominates, versus **manufacturing and retail** where **males** lead. This serves as the foundation for understanding systemic workforce segmentation.

- Top-right, the scatter plot compares education levels with high-income percentages, revealing a **strong positive correlation** between higher education and earnings, but also identifying states where income is **not strictly tied to formal education**. This challenges conventional assumptions and introduces the nuance in upward mobility.

- Below, the heat map provides a geographical lens on income distribution. States like **California, Texas, and Florida stand out for middle-income** density, offering context for economic activity clusters.

- Lastly, the tree map visually breaks down population by **education level and state**, reinforcing which regions hold concentrated education profiles, crucial for contextualizing the trends identified above.

A single-screen layout was intentionally chosen to allow holistic comparison and avoid fragmented insight. Visuals are arranged top to bottom in a logical narrative flow. This sequencing mirrors the analysis journey, guiding the viewer from disparity identification to root cause exploration. The design decisions were purposefully made to reinforce storytelling, ensure readability, and maximize insight clarity for policymakers and socioeconomic planners.